

UD1. VOCABULARY

1. Ciencia vs. Tecnología: La Tecnociencia (Science vs. Technology: Technoscience)

- **Science** (*Ciencia*): The systematic study of the structure and behavior of the physical and natural world through observation and experiment.
- **Technology** (*Tecnología*): The application of scientific knowledge and practical skills to create solutions that meet human needs or solve practical problems.
- **Technoscience** (*Tecnociencia*): The interdisciplinary context where scientific research and technological development are closely linked and mutually dependent.
- **Scientific Method** (*Método Científico*): A rigorous, systematic methodology used to acquire objective knowledge about the universe.
 - **Observation** (*Observación*): Detecting and examining natural phenomena attentively.
 - **Hypothesis** (*Hipótesis*): Proposing a provisional, reasoned explanation for observed phenomena.
 - **Experimentation** (*Experimentación*): Designing tests and measuring quantitative variables to validate or reject a hypothesis.
 - **Conclusion** (*Conclusión*): Determining whether experimental data supports the hypothesis.
- **Knowledge / "To Know"** (*Saber / Conocimiento*): The theoretical understanding of laws, theorems, and scientific models.
- **Execution / "To Make"** (*Hacer / Crear*): The practical action or process of building products, machines, tools, and processes.
- **Natural Law** (*Ley natural*): Immutable principles governing the behavior of the physical universe.

2. El Método Tecnológico y el Proceso de Proyectos (The Technological Process & Project Method)

- **Technological Process / Project Method** (*Proceso Tecnológico / Método de Proyectos*): An ordered sequence of phases used to turn a practical problem into an effective technical solution.

- **Design Constraints / Restraints** (*Condicionantes / Restricciones*): Limitations or requirements imposed on a project, such as budget, safety, materials, dimensions, or time limits.
- **Problem Description** (*Descripción del problema*): Precise formulation of the need or challenge that must be solved.
- **Information Retrieval / Research** (*Búsqueda de información*): The phase dedicated to exploring existing solutions, consulting literature, checking catalogs, or interviewing experts.
- **Ideation & Creativity Techniques** (*Técnicas de ideación y creatividad*):
 - **Brainstorming** (*Lluvia de ideas*): A group creativity technique designed to generate many unfiltered ideas without early judgment.
 - **Mind Mapping** (*Mapa mental*): A visual diagram used to organize ideas, concepts, and relationships around a central theme.
- **Design & Technical Representation** (*Diseño y representación técnica*):
 - **Sketch / Freehand Drawing** (*Boceto*): A quick, initial, freehand drawing of an idea without exact measurements.
 - **Diagram / Blueprint** (*Croquis / Plano*): Detailed technical drawings that specify exact dimensions, components, and geometry.
 - **Perspective / Axonometric View** (*Perspectiva / Vista axonométrica*): 3D representations used to display an object's spatial relationships.
- **Planning & Organization** (*Planificación y organización*):
 - **Bill of Materials (BOM)** (*Lista de materiales*): A detailed table listing all raw materials, components, tools, and estimated costs needed.
 - **Process Sheet / Task Schedule** (*Hoja de proceso / Planificación*): A document outlining step-by-step tasks, assigned team members, estimated durations, and safety guidelines.
- **Prototyping & Construction** (*Prototipado y construcción*):
 - **Prototype** (*Prototipo*): An early operational model built to test, refine, and validate a design concept.
 - **Assembly** (*Montaje / Ensamblaje*): The process of fitting components together according to design specifications.
- **Evaluation & Testing** (*Evaluación y control*):

- **Testing / Trial** (*Pruebas / Ensayos*): Assessing whether the constructed object meets all initial requirements and operates safely.
- **Redesign / Refinement** (*Rediseño / Modificación*): Making adjustments or improvements based on test failures or evaluation outcomes.